

Smart Irrigation System

Presentation Script

CEPDT Project · 4 Speakers · 13 Slides

Each section below belongs to one speaker. Read your part in a natural, conversational tone — you don't need to memorize it word for word, just know the flow.

SPEAKER 1 OF 4

Sohan

Introduction — Slides 1, 2, 3

SLIDE 1 Title

Good morning/afternoon everyone. We are presenting our project — the **Smart Irrigation System**. It's built in two parts: first, a working circuit we designed in Tinkercad, and second, an interactive 3D digital twin we built using Three.js. Let's walk you through both.

SLIDE 2 Project Overview

Our system works in four simple steps. First, it senses soil moisture using a soil sensor and a temperature sensor. Second, we set a custom threshold using a potentiometer — basically a dial that decides how dry the soil should get before watering starts. Third, once that threshold is crossed, the system automatically turns on a water pump, LEDs, and a buzzer. And fourth, all this information is displayed live — both on an LCD screen in the real circuit, and on a 3D screen in our digital model.

SLIDE 3 System Architecture

At the center of everything is the Arduino Uno — think of it as the brain of the system. On one side, it takes inputs: the soil moisture sensor, the temperature sensor, and the threshold dial. On the other side, it controls outputs: the water pump, the LCD display, and the buzzer with LEDs. The Arduino constantly reads the inputs and decides what action to take.

(Hand off to Aayush)

Aayush

Tinkercad Circuit — Slides 4, 5, 6

SLIDE 4**Part 1 Divider**

Now I'll take you through Part One — how we designed and simulated our circuit in Tinkercad, before touching any real hardware.

SLIDE 5**The Circuit**

This is our full circuit, simulated entirely in Tinkercad. It includes the Arduino Uno as the controller, a soil moisture sensor, a TMP36 temperature sensor, a DC motor that acts as our water pump, a 16 by 2 LCD screen for readouts, a potentiometer for setting the threshold, and a buzzer with LEDs for alerts. We tested and verified all the wiring here first, which let us catch mistakes safely before building anything physical.

SLIDE 6**Bill of Materials**

Here is the full list of components we used — every sensor, resistor, and part, along with how many of each we needed. This table is basically our shopping list and wiring reference, all in one place.

(Hand off to Naman)

Naman

Logic + 3D Intro — Slides 7, 8, 9

SLIDE 7**How the Circuit Works**

Here's how the logic actually runs, in four steps. Step one: the sensors sense the soil and temperature. Step two: the Arduino compares that reading to the threshold we set. Step three: it decides and acts — if the soil is too dry, the pump, red LED, and buzzer turn on; if it's fine, the green LED just stays on and the system idles. Step four: everything is displayed live on the LCD screen.

SLIDE 8**Part 2 Divider**

That covers the hardware side. Now let's move to Part Two — where we took this same logic and turned it into a 3D digital twin using Three.js.

SLIDE 9**Inside the Scene**

We built a full 3D farm scene that runs right in a web browser — no installation needed. It includes a sky with weather effects, a water tank, a windmill, a shed, and crop fields. There's also a live HUD, like a dashboard, that shows moisture, temperature, rain, tank level, and system state in real time. Interestingly, none of the textures — grass, soil, wood — are image files; they're all generated by code on the spot.

(Hand off to Manthan)

Manthan

Features + Wrap-Up — Slides 10, 11, 12, 13

SLIDE 10 Play With It

What makes this 3D model special is that it's interactive. You can drag the on-screen dial to change the watering threshold live. You can click a button to simulate dry soil and instantly watch the system respond. You can trigger rain, toggle between day and night, and even orbit the camera around the whole farm by dragging your mouse.

SLIDE 11 Two Builds, One System

So we essentially built the same system twice, in two different ways. The Tinkercad version is a real, hardware-accurate circuit — great for testing and validating before buying parts. The Three.js version is a digital twin — it runs in any browser, needs zero installation, and is perfect for demos and presentations, exactly like this one.

SLIDE 12 Impact & What's Next

This project shows how hardware prototyping and software visualization can come together in one workflow. Looking ahead, we could stream live sensor data from a real board straight into the 3D dashboard, make the system solar-powered for field use, add a mobile app for remote control, and even expand it to manage multiple irrigation zones at once.

SLIDE 13 Thank You

That brings us to the end of our presentation. Thank you for listening — we're happy to take any questions.